Healthcare Dataset Visualization: Operational Trends and Insights from Patients Admissions

Introduction

❖ Objective

This project deploys data visualization to identify trends in healthcare billings, medical conditions and patient admissions.

❖ Its Significance

Better planning, resource allocation, and decision-making result from healthcare professionals' ability to swiftly understand complex data through visualization.

❖ Aim

To illustrate how patient-level data insights can be improved through interactive visualizations.

Dataset Overview

Source: Kaggle (Healthcare Dataset

Records: 55500 rows & Features : 15

Important Fields:

- ❖ "Age", "Gender", " Medical Conditions"
- ❖ "Admission and Discharge Dates"
- ❖ "Bill Amount", "Insurance Provider"
- ❖ Admission Type (Elective, Emergency, Urgent)

Time Frame: Several years of hospital Admission Data.

Purpose: Record hospital operations, treatment trends, and patient demographics.

Data Cleaning & Methodology

Steps for Cleaning:

- ❖ 534 duplicate records were dropped.
- ❖ Standardized formatting performed (Name, Gender, Blood Types)
- ❖ Date features converted to Datetime objects.
- ❖ Negative billings and Invalid hospital stays were filtered.

Feature Engineering:

- ❖ Extracted Length of Stay from admission/Discharge Dates.
- ❖ Created Months and Year from Admission Dates.
- ❖ Text fields were standardized, to help in reliable grouping and standardization.

Tools Used:

The tools that were used are:

- ❖ Python
- ❖ Pandas
- ❖ Seaborn
- ❖ Matplotlib
- ❖ Plotly Express

Feature Relevance for Analysis

❖The primary visual trends were guided by key features such as Age, Gender, Medical Condition, and Admission Type.

❖The purpose of Length of Stay and Billing Amount was to reveal operational and financial effects.

❖Making decisions about patient care and hospital efficiency was improved by concentrating on these

Challenges and Limitations

❖Insufficient Clinical Outcome Information

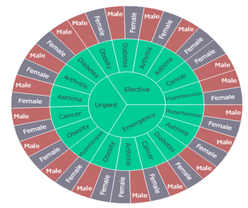
❖Absence of geographic data

❖Possible sampling biases

❖Limited time range

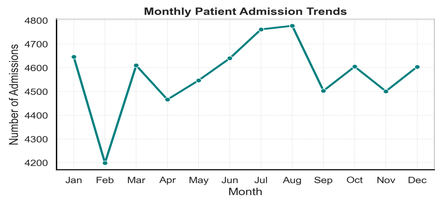
Key Visualization and Insights

Sunburst Chart of Admission Type,Medical Condition, and Gender Distribution



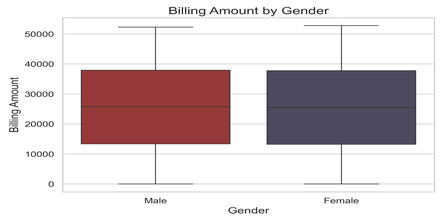
Key Insights:

- ❖ For chronic diseases like diabetes and hypertension, urge nt admissions rule.
- ❖ Gender representation varies women somewhat predominate in diseases including obesity and arthritis.
- ❖ This picture emphasizes the complexity of patient intake and helps to prioritize resources depending on admission urgency.



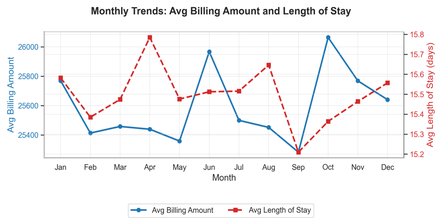
Key Insights:

- ❖ Peak patient admissions in August through October point to seasonal fluctuations in hospital demand.
- ❖ The lowest admission rates in February often point to potential for resource optimization during off-peak times.
- ❖ Aids in hospitals' monthly adjustments of staffing and bed availability.



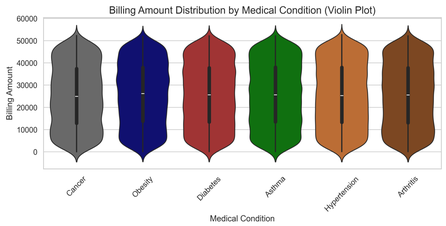
Key Insights:

- ❖ Generally speaking, men and women have similar billing amounts; their distribution differs somewhat.
- ❖ Points up chances for gender-based review of healthcare equity and cost-effectiveness.
- ❖ Guarantees financial plans match patient demographics.



Key Insights:

- ❖ With the highest average billing, October indicated either more extensive treatments or rising service expenses during this time.
- ❖ April's average hospital stay, on the other hand, was the longest, suggesting a possible increase in seasonal or chronic conditions needing more prolonged treatment.
- ❖ Improvements in efficiency during some months could greatly lower running expenses.
- ❖ Promotes improved patient throughput policies and financial planning.



Key Insights:

- ❖ The highest billing variability of cancer and diabetes points to complicated care paths.
- ❖ In diseases like hypertension and asthma, consistent billing patterns point to predictable treatment costs.
- ❖ supports development of insurance packages depending on disease type and budgeting for healthcare.
- ❖ Promotes improved patient throughput policies and financial planning.

Key Insights Summary

- ❖ Mostly for chronic diseases like diabetes and hypertension, urgent patient admissions account for most of hospital intake.
- ❖ Seasonal surges in admissions affect resource allocation and happen between August and October.
- ❖ Different gender patterns of billing and disease frequency affect financial and treatment planning.
- ❖ Concentrated among specific conditions, financial burdens highlight areas of insurance and hospital management priority importance.

Recommendations

❖To handle seasonal admission spikes, increase bed capacity and staffing from August to October.

❖Programs for managing chronic diseases that address obesity, diabetes, and hypertension should be given priority.

❖Conduct financial equity audits to resolve small gender-based billing discrepancies.

❖Reduce hospital stays and billing peaks by streamlining patient discharge procedures.

Implications and Impacts

Hospital managers can use the study's insights to better allocate resources, plan finances, and create specialized treatment plans for patients with chronic illnesses. Modifying coverage plans in response to billing variability across conditions can also be advantageous for insurance companies. All things considered, data-driven tactics can improve patient outcomes and healthcare facilities' operational effectiveness.

Conclusions

This project effectively illustrated how healthcare datasets can reveal hidden patterns through interactive data visualizations. Important trends in patient admissions, billing amounts, and medical conditions were identified through thorough data cleaning, feature engineering, and the use of both simple and sophisticated visualization techniques. The results provide insightful information to aid in financial decision-making, resource allocation, and healthcare planning. The project emphasizes the vital role that visual storytelling plays in turning raw data into actionable intelligence, despite the dataset's limitations. To gain deeper insights, future research can build on this foundation by integrating clinical outcomes and geographic diversity.

References

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